

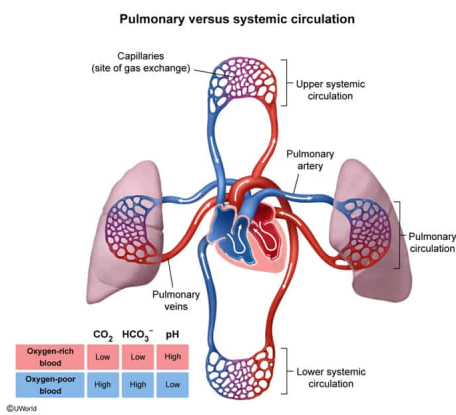
Compared with blood in the pulmonary veins, blood in the pulmonary arteries has higher concentrations of:

- I. O_2
 - II. H^+
 - III. CO_2
-
- ☐ A. I and II only [20%]
 - ☐ B. I and III only [3%]
 - ✓ ☒ C. II and III only [75%]
 - ☐ D. I, II, and III [1%]

Correct

74% answered correctly

Explanation:



Blood is circulated throughout the body by the repetitive contraction of the heart, which enables continuous delivery of oxygen and nutrients to the body's tissues, as well as removal of CO_2 (the major metabolic by-product of [cellular respiration](#)) and other waste products. The blood vessels involved in circulation can be divided into two circuits:

- The **systemic circuit** distributes **oxygenated** blood (high O_2 , low CO_2 , low H^+) containing nutrients to all body tissues via **arteries**, which carry blood away from the heart. When oxygenated blood reaches the capillaries within the systemic tissues, it **delivers** O_2 (becomes deoxygenated) and nutrients to the tissues and picks up CO_2 . **Veins** then return the **deoxygenated** blood (low O_2 , high CO_2 , high H^+) back to the heart so that it can enter the pulmonary circuit.
- The **pulmonary circuit** pumps **deoxygenated** blood through the **pulmonary arteries** to capillaries in the lungs to facilitate [gas exchange](#). The capillaries are in close contact with lung alveoli (air sacs), and at this site, the blood becomes reoxygenated (O_2 is transferred from alveolar air to the blood, and CO_2 is transferred from the blood to the alveolar air). The **pulmonary veins** then return the **oxygenated** blood to the heart so that it can be pumped through the systemic circuit (**Number I**). Compared with oxygenated blood in the pulmonary veins, deoxygenated blood in the pulmonary arteries has *higher* concentrations of CO_2 (**Number III**).

Blood that is richer in CO_2 is also generally richer in H^+ ions due to the following reaction that takes place in red blood cells:



The higher the CO_2 concentration, the more the forward reaction is favored. Therefore, the CO_2 -rich blood in the pulmonary arteries also contains *more* H^+ ions (has a slightly lower pH) than the CO_2 -poor blood of the

pulmonary veins (**Number II**).

Educational objective:

Within the pulmonary circuit, the pulmonary arteries carry CO₂-rich, O₂-poor blood from the heart to the lungs, where CO₂ is exchanged for O₂. The pulmonary veins then carry CO₂-poor, O₂-rich blood from the lungs to the heart, which then pumps the oxygenated blood to all body tissues.

Time spent:0 sec

Subject:
Biology

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System:
Circulation and Respiration